PROPOSIFY

SAI2 Project

Technical Report

[GitHub]

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1. Introduction

The goal of the project is to support BFH students during the early stages of their bachelor thesis by offering a reliable and academically aligned tool. Instead of replacing the student's work, the assistant is designed to guide them through the process, help them clarify their ideas and make the proposal phase easier for the students.

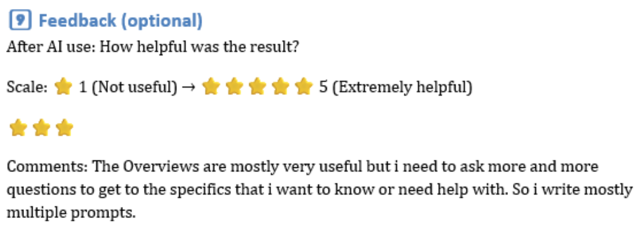
In the following chapters, we will describe the design decisions, technical architecture and implementation details that shaped the system. The report outlines how the different components work together and provides an overview of the assistant's structure and constraints.

2. Requirements & Specifications

2.1 Problem Understanding

To define the requirements of our thesis proposal assistant, it was important for us to first understand how users, which in our case were students, currently formulate their inputs and where they usually encounter difficulties. That's why we decided to conduct 9 user interviews among the students. We created a word form with all questions and information that we needed to gather from the users [see User Prompts]. After reviewing the word forms, it became obvious that most of students start with very rough and unstructured ideas.

Many students only write down a broad topic without really knowing yet how to turn that into a solid research question or a proper proposal structure. Others say they don't know what the next step should be or how to spot a research gap that is worth exploring. Even those who seem a bit more advanced often still need help refining their question or choosing a method.



9 Feedback (optional)

After AI use: How helpful was the result?

Scale: ⭐ 1 (Not useful) → ⭐⭐⭐⭐⭐ 5 (Extremely helpful)

⭐⭐⭐

Comments: The Overviews are mostly very useful but i need to ask more and more questions to get to the specifics that i want to know or need help with. So i write mostly multiple prompts.

FIGURE 1 - FEEDBACK FROM USER FORM

The forms also show that students already use generic AI but often receive answers that are too broad or they need to repeatedly refine prompts to obtain specific information [Figure 1 - Feedback from user form]. This shows us a clear problem - existing tools do not adapt to the student's progress and can end up creating confusion instead of providing clarity.

Another important finding is students preparation level. Some have no research question at all, others provide partial drafts and a few already outline methodological ideas. Because of this, the assistant must adapt to different levels of draft maturity and respond appropriately based on the information provided.

Our assistant needs to support students in turning their ideas into clear research questions, provide a structure of the proposal and give reliable guidance without hallucinating sources. It must adapt to different stages of student progress, whether they have only a topic, a partial draft or a nearly finished proposal and offer help with refining scope, identifying research gaps and selecting suitable methods.

2.2 Solution understanding

This section defines what an appropriate solution for the identified problem should achieve, based on insights from five expert interviews.

2.2.1 Expert Perspective and Role

The interviewed experts are thesis supervisors who regularly evaluate bachelor thesis proposals and support students in the early research phase. Through their experience, they have a clear understanding of common student difficulties, institutional requirements, and the criteria used to assess proposals.

Proposal writing is a complex task in which students have to define a problem, engage with literature, and design a feasible research approach, often with limited prior experience (Wentz, 2014). All experts agreed that students often struggle with unclear or overly broad research questions, weak alignment between the research question and methodology, unrealistic project scope, and limited academic structure and critical thinking. These recurring problems define the key areas that an effective solution must address.

2.2.2 Ideal Experts Support

The ideal outcome is not a finished proposal, but a student who has gained clarity and confidence. Success means that students leave the proposal phase with a feasible research question, an understanding of its relevance, and a methodology they can explain and justify.

Proposal phase should help students develop a realistic and defensible plan rather than produce a perfect document. Students should also feel less overwhelmed and have realistic expectations about the workload and scope of a bachelor thesis.

2.2.3 Gold Standard for Proposals and AI-Supported Outcomes

Experts described high-quality proposals as clearly structured, coherent, and realistic in scope. Strong proposals explain the relevance of the topic, identify a research gap, formulate an answerable research question, and justify the chosen methodology. Within this gold standard, experts clearly defined the role of AI. AI cannot replace literature search and evaluation, as this requires academic judgment. However, AI can support students by helping them structure ideas, reflect on their reasoning, and summarize key insights from provided materials. Effective results therefore depend on student-provided content combined with AI-supported guidance.

3. Prototype Design

This chapter describes how the insights from the problem and solution understanding were translated into concrete design decisions.

3.1 Design Overview

At the start of the project, many aspects of a chat-based thesis proposal assistant were still unclear. Students already use existing AI tools (including document-chat systems where papers can be uploaded), but these tools are mostly generic and not tailored to BFH requirements, structured supervision, or clear academic guidelines.

During early discussions and experiments, Retrieval-Augmented Generation (RAG) emerged as a promising basis, as meaningful structural guidance requires access to reliable academic sources. In particular, the BFH thesis guidelines and Creswell's *Research Design* book were identified as key references, as they reflect institutional expectations and established research practices. However, it soon became clear that a purely RAG-based solution would be too rigid. Constant retrieval for every user input is unnecessary and does not support reasoning, reflection, or creative exploration.

To structure these design decisions, we refer to the design space for intelligent and interactive writing assistants proposed by Wambsganss et al. (2024), as presented in Figure 2

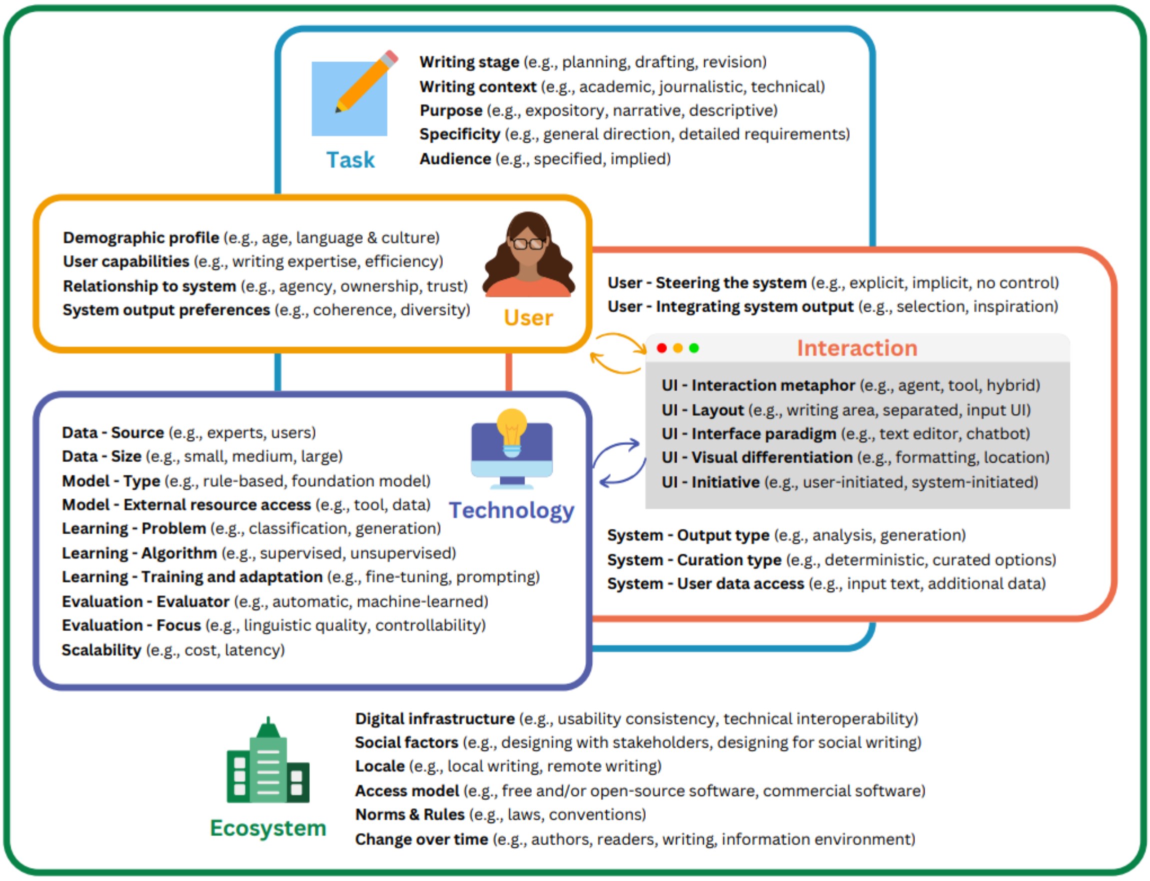


FIGURE 2: DESIGN SPACE FOR INTELLIGENT AND INTERACTIVE WRITING ASSISTANTS (FROM WAMBSGANSS ET AL., 2024, P. 5).

The framework describes writing assistants along dimensions such as task stage, user role, interaction style, technology, and ecosystem constraints. We used it as a reference rather than a system to be implemented. Our prototype focuses on thesis proposal work, specifically the planning stage of writing the paper. It supports students by structuring ideas, asking clarifying questions, and suggesting next steps, instead of generating complete proposal text.

As a result, the design focus shifted toward a system that can decide when and how retrieved knowledge should be used. The assistant was therefore implemented as an agent-based system rather than a single static prompt. LangGraph was chosen because it allows explicit control over reasoning steps and decision flows.

The structure of the agent was strongly influenced by the expert interviews. Two main areas were identified where students need the most support: developing a strong research question and writing a coherent thesis proposal. These tasks differ in reasoning style and use of sources, which led to two reasoning flows. One flow focuses on research-question development and encourages students to engage with relevant literature before generating or refining research questions, while still allowing controlled creativity. The second flow supports proposal writing and refinement and is grounded in the official BFH proposal template. Due to privacy constraints, publicly available pre-registered research proposals from the Open Science Framework were used as reference examples.

From a user interaction perspective, the prototype was implemented as a Streamlit web application. Students select a persona, an AI model, and an answer style before starting, reflecting different preferences regarding tone and level of critique.

3.2 System architecture

This section describes the overall architecture of the thesis assistant and explains how the system combines user interaction, agent orchestration, retrieval, and language model generation.

3.2.1 Overall Architectural Structure

At runtime, user input is processed centrally and routed to the appropriate agent workflow. The agents decide whether retrieval is required, which knowledge sources should be used, and how retrieved material should be integrated into the prompt before calling the language model through a dedicated service layer.

3.2.2 Knowledge Sources and Grounding Strategy

The assistant relies on two main categories of knowledge, which are treated differently to ensure both reliability and relevance.

First, the system uses a static academic knowledge base containing methodological and institutional guidance, including BFH recommendations and material derived from Creswell's *Research Design*. This content is stored in a vector database (ChromaDB) to enable semantic retrieval.

Second, the system incorporates student-provided literature in the form of uploaded PDF documents. These PDFs are not directly retrieved chunk-by-chunk during every query. Instead, they are converted into structured paper summaries (e.g., topic, research question, methodology, data, findings, and gaps). These summaries are stored in the session state and reused across turns. This design reflects the insight that AI can only provide meaningful structure when the relevant content is available and supplied by the student. A limitation of this approach is that very specific or fine-grained questions about an individual paper may not be answerable if the corresponding details are not included in the summary. In such cases, students are explicitly advised to consult the original paper. This design decision was made to avoid overloading prompts and context windows and to instead prioritize high-level information such as the research question, methodology, data, key findings, and limitations. These elements are particularly relevant for shaping a student's own research question and methodological choices during the proposal phase.

The LangGraph agents decide what to retrieve and how to incorporate these sources depending on the task. Not every user message requires retrieval, for many interactions, the system can respond based on conversational context and student-provided information alone.

3.2.3 Retrieval-Augmented Generation

This section describes the RAG subsystem, including embedding creation, vector search, and how retrieved knowledge is combined with the language model to produce grounded answers.

Embeddings, ChromaDB, and Ollama

Semantic retrieval in the system is based on embeddings. When the system needs to retrieve relevant guidance from the static knowledge base, the user's question (or a refined retrieval query) is converted into a high-dimensional vector representation. The underlying principle is that semantically similar texts are mapped to nearby points in vector space, enabling semantic search beyond keyword matching. The system uses the following components:

- **Embedding computation (Ollama):** Embeddings are generated locally via an Ollama embedding endpoint.

- **Vector storage and search (ChromaDB):** ChromaDB stores pre-computed embeddings of knowledge base chunks and supports nearest-neighbor search. For a query, ChromaDB returns the most similar chunks along with their metadata.

- **Retrieval logic:** A retrieval function embeds the query, searches the vector database, and returns the most relevant text passages. When needed, retrieval can also prioritize BFH-specific sources to ensure alignment with institutional expectations.

3.2.4 Research question pipeline

When the application is used in "Research question helper" mode, each user message is processed through the research-question workflow, implemented as a LangGraph state. Figure 3 illustrates the routing logic and execution paths of this workflow.

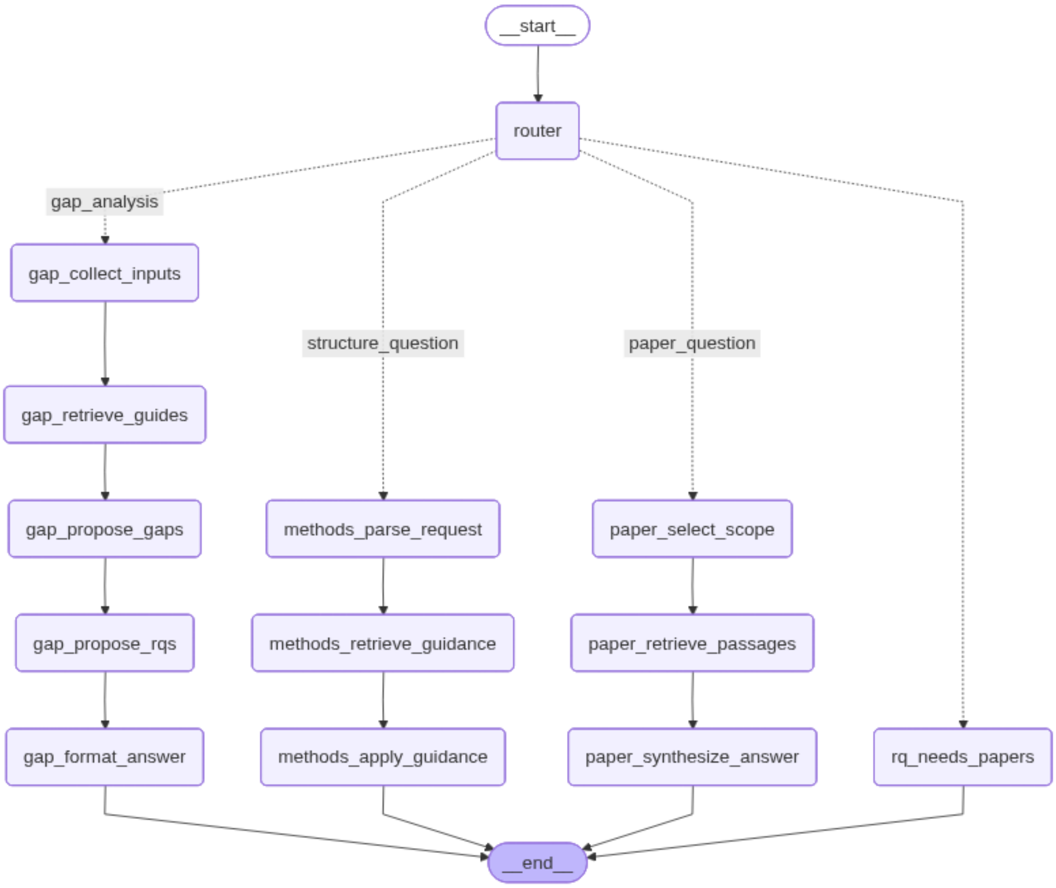


FIGURE 3 RESEARCH QUESTION LANGGRAPH WORKFLOW

Step 0: Input and State Setup

Each user turn begins in the Streamlit application. The central handler constructs the initial graph state (AppState), which includes the student's current question, the selected mode and persona, a compact session summary, a compressed representation of recent interactions, and structured summaries of any uploaded PDFs. This state is passed to the LangGraph workflow and serves as the shared memory across all subsequent reasoning steps.

Step 1: High-Level Routing

The first node in the graph is a router that uses the language model as a lightweight classifier. For each user input, it assigns exactly one task label, which determines the execution path of the graph. The possible routing outcomes are:

- **structure_question:** questions related to research-question quality, research design, or methodological choices.
- **gap_analysis:** requests to identify research gaps and propose candidate research questions grounded in the uploaded literature.
- **paper_question:** questions about the content of one or more uploaded research papers.
- **rq_needs_papers:** requests for research-question generation when insufficient literature has been provided.

This routing step ensures that the assistant selects an appropriate reasoning strategy instead of applying retrieval or brainstorming uniformly to all inputs.

Path A: Methods and Research-Question Refinement (structure_question)

When the router selects **structure_question**, the pipeline focuses on improving the clarity, scope, and feasibility of the student's research question and its alignment with suitable methods. The system first interprets the student's intent (e.g., critique an existing question, propose a design, or refine wording). It then retrieves relevant methodological guidance from the static BFH/Creswell knowledge base using semantic search.

The final response is generated by combining the student's question, the compact session memory, retrieved guidance, and persona instructions into a structured prompt. The output provides targeted feedback or refinement suggestions and explicitly links the research question to feasible methodological choices.

Path B: Paper-Based Questions (paper_question)

When the router selects **paper_question**, the pipeline answers questions that explicitly refer to one or more uploaded research papers. The system first determines which papers are relevant to the question, for example by matching filenames or references mentioned by the student. It then retrieves the corresponding structured paper summaries from the session state and prepares them as context.

Using these summaries and the student's question, the language model generates a focused, paper-grounded answer.

Path C: Gap-Based Research-Question Generation (gap_analysis)

If the router selects **gap_analysis**, the pipeline shifts to literature-driven reasoning. All available paper summaries are consolidated into a structured context, preserving one section per paper. In parallel, additional guidance on research gaps and contribution formulation is retrieved from the static knowledge base.

Based on these inputs, the language model is asked to propose a small set of plausible research gaps and subsequently transform the strongest gaps into concrete candidate research questions, each accompanied by a short justification.

Path D: Insufficient Literature Provided (rq_needs_papers)

If the router selects **rq_needs_papers**, the system does not attempt to generate research questions. Instead, it returns a short guidance message indicating that too few papers have been provided and instructs the student to upload and summarize additional literature before continuing. This enforces the intended usage pattern that research-question generation should be grounded in a minimal literature base.

3.3.5 Proposal Refinement Pipeline

We implemented the proposal refinement assistant as a separate LangGraph flow because its purpose is quite different from research-question assistant. This pipeline is designed to help students improve and refine proposal drafts that they have already started. The flow begins with a router node that decides whether the student is asking for general guidance or wants an actual refinement of pasted text.

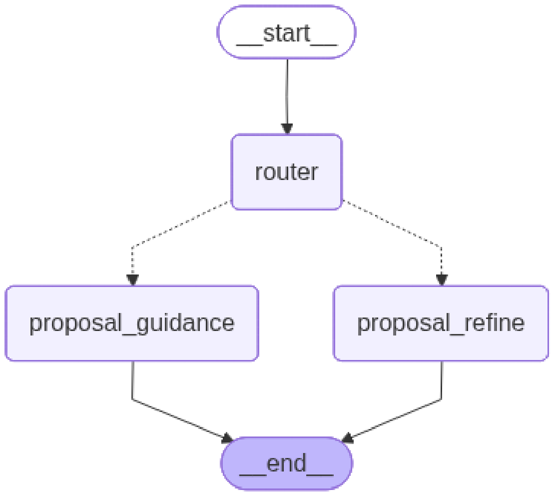


FIGURE 4 PROPOSAL REFINEMENT LANGGRAPH WORKFLOW

As can be seen in Figure 4, the flow starts with a router node that classifies each user message as either guidance (general proposal advice) or refine (student-pasted draft text). The router first checks for long, structured input or BFH template keywords, if these are present and user context exists, it directly selects the refine path. Otherwise, a classifier prompt (temperature = 0, GPT-OSS 120B) determines the route.

Path A: Proposal guidance assistance

In **proposal_guidance** mode, the assistant retrieves approximately five chunks from the Creswell and BFH knowledge base, trims the context to about 2.5k characters and combines it with persona instructions, the session summary and recent Q&A. The response is limited to around 1,800 tokens.

Path B: Proposal refinement assistance

In **proposal_refine** mode, the pasted draft is shortened before retrieval, six context chunks are fetched with BFH guidelines. Then the BFH proposal template outline, uploaded paper summaries, specialization examples and persona settings are injected into the prompt. As part of the proposal refinement process, the system includes specialization-specific thesis examples provided as Markdown files. These examples correspond to different BFH specializations (e.g. Applied Artificial Intelligence, Business Data Analytics, Software Architecture), we gathered those examples from real thesis papers from the internet. Once a specialization is selected, a short snippet of the corresponding example is injected into the system prompt. The refinement output is also capped at approximately 1,800 tokens. It highlights any modified lines and includes a short source note. After execution, temporary metadata and context are cleared to keep the next turn clean.

3.2.6 Memory and Context Handling

All state is stored in the Streamlit session, including chat history, rolling summary, recent sources, selected mode, persona, model, uploaded paper summaries and the last task. Before each response, a recent-history window is rebuilt with a limit of 1,800 tokens (tiktoken cl100k_base), going backwards through the conversation. After each turn, a short session summary is refreshed using update_summary_ephemeral shown in Figure 5

```python
def update_summary_ephemeral(history, current_summary: str) -> str:
    chat_turns = [h for h in history if h.get("kind", "chat") == "chat"]
    pairs = []
    for item in chat_turns:
        u = _clean_text(item.get("frage", ""))
        a = _clean_text(item.get("antwort", ""))
        if u or a:
            pairs.append(f"USER: {u}\nASSISTANT: {a}")
    recent_text = "\n\n".join(pairs) if pairs else "None"
    prompt = f"""You are a thesis-proposal assistant's memory summarizer.

Rules:
- Output at most 5 bullet points.
- No checklists, no meta-instructions, no quotes from the chat.
- Keep only: current research question, scope/constraints, chosen methods/datasets, deadlines/supervisor notes, key decisions & open TODOs.
- No rhetorical prompts and any 'self-reflection' text.

[Existing summary]
{_clean_text(current_summary)}

[Conversation turns to merge]
{recent_text}

[Write the updated summary as 3-5 bullets only, no header:]
"""
    try:
        new_summary = llm_complete(prompt, max_tokens=180, temperature=0.0)
        return _clean_text(new_summary) or current_summary
    except Exception:
        return current_summary
```

FIGURE 5 UPDATE_SUMMARY_EPHEMERAL CODE FUNCTION

This function summarizes the conversation into a small set of bullet points. Uploaded paper summaries are cached per session, retrieval queries are truncated to approximately 1.2k characters and outputs are capped at around 1,800 tokens. Safety checks prevent the assistant from generating full thesis sections and keep the interaction focused on guidance and refinement.

4. Prototype Deployment

The prototype was implemented as a containerized application using Docker and Docker Compose. This approach was chosen to keep the system modular, reproducible, and easy to run. Instead of running all components in a single process, the system was split into three separate services: the Streamlit application, a vector database, and a local embedding service.

The main application runs inside a custom Docker image and hosts the Streamlit web interface. This container includes all required Python libraries, such as LangGraph for agent logic, the Chroma client for retrieval, and PyMuPDF for processing uploaded PDFs. It exposes the Streamlit interface and acts as the central component that connects user interaction with retrieval and language model calls.

A second container runs ChromaDB, which is used to store embeddings for the static academic knowledge base. This includes BFH guidelines and methodological guidance based on Creswell's *Research Design*. The database is accessed through an HTTP interface and uses a persistent volume so that embeddings remain available across restarts.

The third container runs Ollama and is used only for embedding generation. When the application needs to retrieve relevant guidance, it sends text to the Ollama service, which returns vector embeddings. If the required embedding model is not yet available, Ollama downloads and stores it automatically.

Deployment Considerations

The current setup is mainly intended for local use and prototyping. However, because all components are containerized, the system can also be deployed in other environments with limited changes. For example, the same setup could be run on a cloud-based virtual machine or adapted for a container orchestration platform.

5. Prompt Engineering

Prompt engineering is a core part of the thesis assistant, as the system is designed to guide students rather than generate complete thesis content. To ensure controlled and consistent behavior, the assistant uses structured prompt templates instead of a single generic prompt. These templates are divided into global prompts and task-specific prompts and combine zero-shot and few-shot prompting.

5.1 Global Prompts

Global prompts define the assistant's general behavior and apply to all interactions. Each prompt includes a shared safety and grounding preamble that instructs the model to avoid hallucinated sources and not generate complete proposal sections. In addition, global prompts include mode- and persona-specific instructions, which adjust the tone and level of critique without affecting academic content.

5.2 Task-Specific Prompts

Task-specific prompts are used to guide the assistant's behavior for different types of user requests. Separate prompt templates exist for tasks such as research-question critique, method guidance, gap analysis, answering questions about uploaded papers, and proposal refinement. Each task-specific prompt explicitly describes the expected output structure, for example by requesting short paragraphs, bullet points, or section-by-section feedback.

5.3 Zero-Shot Prompting

For many tasks, zero-shot prompting is sufficient. In these cases, the assistant receives only task instructions and relevant context, such as retrieved academic guidance or student-provided paper summaries. Zero-shot prompts are mainly used for analytical tasks, including research-question critique, methodological reasoning, and gap identification, where flexibility and reasoning are more important than stylistic consistency.

5.4 Few-Shot Prompting

Few-shot prompting is used selectively for tasks where output structure and style are particularly important, most notably in proposal-related guidance and refinement. In these cases, the system prompt has more structured sections, including task instructions, persona tone, retrieved contextual knowledge and short example snippets.

In this condition we additionally give the model a short snippet from a real proposal example. These examples correspond to different BFH specializations and are stored in a specified folder in our repository. The snippet is used only as a style and structure guide (headings, level of detail, tone), not as content to copy, helping the model understand how proposals in that specialization are typically organized and written. The whole process of prompt engineering is documented in the notebook (ragchatbot/prompt_comparisons.ipynb).

6. Evaluation

After we finished our prompt engineering and prototype, we again conducted a couple of short interviews with students. To evaluate the prototype, we asked students from BFH to test the application and share their impressions. We performed short, casual interviews and let the students use our app freely and comment on it. Our goal was to understand how usable the system is, how students feel about its outputs and where confusion usually occurs during interaction.

6.1 User insights

Overall, the prototype created a positive first impression. Most students were able to start using the tool immediately without additional explanation.

Students were positively surprised by the depth and structure of the responses. Instead of generic answers, the assistant provided concrete guidance, proposal structure suggestions, and next steps. Several students noted that the tool felt more useful than generic AI systems because it was clearly aligned with academic proposal writing and BFH expectations.

However, we also identified some weaknesses. Users occasionally felt confused about the different agent modes, as the distinctions between them were not always visually clear. This sometimes led to mismatched expectations, particularly for literature-based gap analysis. In response, additional help buttons and explanation boxes were added.

Another issue concerned PDF handling. Some students were unsure whether their documents were uploaded correctly or whether answers were based on their papers or general knowledge. Missing or unclear references reduced trust in these cases.

In summary, the evaluation shows that the core concept works well, but clearer feedback about system behavior and data usage is needed to further improve usability and trust.

6.2 Iteration and Refinement

Based on user feedback, we managed to identify different improvements. First, agent selection should be clarified. Now, our prototype has larger, more distinct selection cards and stronger highlighting of the active agent that help students immediately understand which mode they are in. Second, the PDF upload and processing flow needs clearer feedback. The system now explicitly indicates when a document has been uploaded, when it is being processed and when it is ready for use. Showing document names and clarifying that multiple PDFs are supported.

Finally, when answers are informed by uploaded papers, explicit citation information should be included. We couldn't resolve this issue in time, adding a citation option with exact page numbers and papers were out of scope for us, so we decided to keep it as a future improvement.

In summary, the evaluation confirms that the prototype meets its core goals while highlighting refinements needed to better support students in ways that are consistent with academic expectations and BFH paper guidelines.

7. Conclusion & Discussion

The goal of this project was to build a thesis proposal assistant that supports BFH students during the early proposal phase without replacing the learning process. Based on user and expert interviews, the prototype was designed to act more like an academic supervisor than a generic text generator. Instead of producing full proposal sections, the assistant focuses on clarifying questions, challenging weak assumptions, suggesting realistic next steps, and keeping responsibility for content decisions with the student.

The qualitative evaluation with students indicates that the prototype meets these goals. Students described the system as approachable, academically aligned, and more useful than generic AI tools. In particular, they valued the structured outputs, such as topic overviews, proposal guidance, and concrete next steps, which helped them understand how to proceed rather than simply what to write.

The overall concept and structure of the system worked well. Splitting support into research-question development and proposal refinement reflects how students typically progress during the proposal phase and helps keep feedback focused and manageable. This follows the idea that complex research tasks become more approachable when broken into smaller steps (Turabian, 2018, p. 20).

From a technical perspective, the combination of agent-based workflows, Retrieval-Augmented Generation grounded in BFH guidelines and Creswell, and structured handling of student-provided literature proved effective in delivering relevant and academically grounded guidance.

Prototype limitations

Main limitations relate to privacy and the handling of academic materials. During development, the prototype relied on institutional guidelines, research methods literature, and proposal-style examples to learn structure and tone. However, using real student proposals or internal supervisor examples was not possible.

A second limitation is that the assistant can only improve proposal quality if students provide enough input, such as draft text, constraints, and relevant literature. A proposal must clearly state the problem, explain its relevance, show how others have approached it, and describe how the student plans to proceed (Wentz, 2017, pp. 15–16). For this reason, the prototype does not aim to generate a full proposal. Instead, it focuses on helping students set boundaries, improve feasibility, and make better decisions based on the material they provide.

References

Turabian, K. L. (2018). *A manual for writers of research papers, theses, and dissertations* (9th ed.). University of Chicago Press. https://www.joeteacher.org/uploads/7/6/3/0/7630382/turabian_manual_9th_ed.pdf

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Appendix

Problem Understanding Survey

The survey was conducted in the early stage of the project. The survey collected responses from 8 students working on a bachelor thesis proposal. It shows student goals, challenges and expectations related to thesis proposal writing. It also shows that students mainly struggle with defining a clear topic, planning the project, and choosing suitable methods for the proposal.

User Prompts

[Link to the survey]

Student 1: Cyril

Basic Information

Study Program: Business IT
Specialization: Software Design & Architecture

Thesis Topic

Briefly describe your intended topic.

Standardize and Improve Portfolio Epic Flow in a Business Unit. Follow up of Praxisprojekt where the Portfolio Epic Flow was standardized and improved for a Cluster

Organizational Diagram

CTO -> Business Unit -> Cluster -> DevOps Team

Research Question(s)

Write your main research question(s).

Develop a Recommendation based on the current state of the art and the literature internal of the company and existing literature (f.e. SAFe, Lean Management, Scrum etc.)

Background Knowledge

How much do you already know about this topic?

☐ Nothing yet
☐ Some general understanding
☐ Moderate knowledge (read some papers/articles)
☒Deep knowledge (previous coursework, job experience)

(Optional: briefly describe your current understanding.)

Goal of the AI Prompt

What do you want AI to help you with? (multiple allowed)

☒Formulating or refining my research question

☒Creating a structure or outline
☒Defining methodology or data sources
☒Writing summaries or theoretical background
☒Clarifying concepts or terminology
☒Generating visualizations or project plans
☐ Something else: ______________________________

Actual Prompt for AI

Write your best attempt at the prompt you'd like to give the AI.

- Uploaded notes from meetings with client from work and asked to map it to the proposal and then added further inputs where was needed.
- Example:

I want to write my thesis on AI and copyright law. Please help me define 3 possible research questions that are specific, measurable, and relevant to Swiss or EU regulations.

Reorganization until December 2025

Expected Outcome

What kind of result would you like AI to produce from your prompt?

☒A refined research question
☒A detailed outline or structure
☒A summary or theoretical framework
☒A list of datasets or tools
☒A project plan (Gantt chart, milestones)
☐ Other: ______________________________

Feedback (optional)

After AI use: How helpful was the result?

Scale: ⭐ 1 (Not useful) → ⭐⭐⭐⭐⭐ 5 (Extremely helpful)

Comments: ______________________________

Student 2: Carmelo

Basic Information

Study Program: Business IT
Specialization: Applied artificial intelligence

Thesis Topic

Briefly describe your intended topic.

- Human-centered AI for Education

Research Question(s)

Write your main research question(s).

- How does AI influence the learning process and how does the performance of the students change

Background Knowledge

How much do you already know about this topic?

☐ Nothing yet
☐ Some general understanding
☒Something inbetween
☐ Moderate knowledge (read some papers/articles)
☐ Deep knowledge (previous coursework, job experience)

(Optional: briefly describe your current understanding.)

Goal of the AI Prompt

What do you want AI to help you with? (multiple allowed)

☒Formulating or refining my research question
☐ Creating a structure or outline
☒Defining methodology or data sources
☐ Writing summaries or theoretical background
☐ Clarifying concepts or terminology
☐ Generating visualizations or project plans
☐ Something else: deep-searches into topic to find more research paper

Actual Prompt for AI

Write your best attempt at the prompt you'd like to give the AI.

I want to write my bachelor thesis on Human-Centered AI in Education. Please help me define 3 possible research questions that are specific, measurable, and relevant to the use of AI systems in Swiss or European educational contexts. The questions should focus on how AI can support learners or teachers while ensuring ethical use, transparency, and trust in educational environments.

Expected Outcome

What kind of result would you like AI to produce from your prompt?

☐ A refined research question
☒A detailed outline or structure
☒A summary or theoretical framework

☒A list of datasets or tools
☐ A project plan (Gantt chart, milestones)
☐ Other: ______________________________

Feedback (optional)

After AI use: How helpful was the result?

Scale: ⭐ 1 (Not useful) → ⭐ ⭐ ⭐ ⭐ ⭐ 5 (Extremely helpful)

Comments: ______________________________

Student 3: Stettler Gil

Basic Information

Study Program: Business IT
Specialization: Software Design & Architecture

Thesis Topic

Briefly describe your intended topic.

- UX-refinement of Me and My Planet (beta version of a footprint calculator)

Research Question(s)

Write your main research question(s).

- How to apply different UX-methods to gather human-centered insights and refine an existing User Interface.

Background Knowledge

How much do you already know about this topic?

☐ Nothing yet
☒Some general understanding
☐ Moderate knowledge (read some papers/articles)
☐ Deep knowledge (previous coursework, job experience)

(Optional: briefly describe your current understanding.)

Goal of the AI Prompt

What do you want AI to help you with? (multiple allowed)

☒Formulating or refining my research question
☒Creating a structure or outline
☐ Defining methodology or data sources

☐ Writing summaries or theoretical background
☐ Clarifying concepts or terminology
☐ Generating visualizations or project plans
☐ Something else: ______________________________

Actual Prompt for AI

Write your best attempt at the prompt you'd like to give the AI.

I'll write a Bachelor thesis on how to apply UX-Methods and how to evaluate the insights to propose changes on the existing UI. Give me a few examples on how to formulate my research question and propose a possible structure for the paper.

I want to write my thesis on AI and copyright law. Please help me define 3 possible research questions that are specific, measurable, and relevant to Swiss or EU regulations.

Expected Outcome

What kind of result would you like AI to produce from your prompt?

☒A refined research question
☒A detailed outline or structure
☐ A summary or theoretical framework
☐ A list of datasets or tools
☐ A project plan (Gantt chart, milestones)
☒Other: _based on my input____________________________

Feedback (optional)

After AI use: How helpful was the result?

Scale: ⭐ 1 (Not useful) → ⭐ ⭐ ⭐ ⭐ ⭐ 5 (Extremely helpful)

Comments: ______________________________

Student 4: Thomas

Basic Information

Study Program: Business IT
Specialization:

Thesis Topic

Briefly describe your intended topic.

- Not defined yet, But i would like to develop some kind of AI or Machine Learning System

Research Question(s)

Write your main research question(s).

- Not defined yet, But i would like to test wether AI can be used to assist a specific topic im interessted in.

BACKGROUND KNOWLEDGE

How much do you already know about this topic?

☒Moderate knowledge (read some papers/articles)

I have a bit of experience with developing AI and Machine Learning Systems from multiple Data analysis and AI courses in school. We trained multiple predictiv models and uses LLM's for chatbots.

GOAL OF THE AI PROMPT

What do you want AI to help you with? (multiple allowed)

☐ Formulating or refining my research question
☐ Creating a structure or outline
☐ Defining methodology or data sources
☐ Writing summaries or theoretical background
☐ Clarifying concepts or terminology
☐ Something else:

ACTUAL PROMPT FOR AI

Write your best attempt at the prompt you'd like to give the AI.

I want to write my proposal for my bachelor thesis as a student in business information system in switzerland. I focus in applied artificial intelligence and data analytics for my studies. I would like to develop something in my thesis. What different options do i have for my thesis. What do i need to consider for my proposal. Can you give me some ideas for topics?

I start with some general information about what i want to do and what the context is.
I mostly start from the top down to get some general information, get an overview so i can see wheter there is something i have not considered myself and then proceed to ask more and more specific questions till i get down to specifics.

EXPECTED OUTCOME

What kind of result would you like AI to produce from your prompt?

☐ A refined research question
☐ A summary or theoretical framework
☐ Other: Overview and general information gathering

FEEDBACK (OPTIONAL)

After AI use: How helpful was the result?

Scale: ⭐ 1 (Not useful) → ⭐⭐⭐⭐⭐ 5 (Extremely helpful)

⭐⭐⭐

Comments: The Overviews are mostly very useful but i need to ask more and more questions to get to the specifics that i want to know or need help with. So i write mostly multiple prompts.

Student 5: Lubna

Basic Information

Study Program: **Business IT**
Specialization: **Project Management & Agility**

Thesis Topic

Briefly describe your intended topic.

Research Question(s)

Write your main research question(s).

- Example: To what extent can AI assist in detecting license conflicts in open-source repositories.
- Online Trust & International Communications

Background Knowledge

How much do you already know about this topic?

☐ Nothing yet
☐ Some general understanding
☒Moderate knowledge (read some papers/articles)
☐ Deep knowledge (previous coursework, job experience)

(Optional: briefly describe your current understanding.)

Goal of the AI Prompt

What do you want AI to help you with? (multiple allowed)

☐ Formulating or refining my research question
☐ Creating a structure or outline
☐ Defining methodology or data sources
☐ Writing summaries or theoretical background
☐ Clarifying concepts or terminology
☐ Generating visualizations or project plans

☐ Something else: ______________________________

Actual Prompt for AI

Write your best attempt at the prompt you'd like to give the AI.

I would like to write my bachelor thesis on online Trust and International Communications. Could you help me define 3 research questions that are relevant for this topic.

Expected Outcome

What kind of result would you like AI to produce from your prompt?

☐ A refined research question
☐ A detailed outline or structure
☐ A summary or theoretical framework
☐ A list of datasets or tools
☐ A project plan (Gantt chart, milestones)
☐ Other: ______________________________

Feedback (optional)

After AI use: How helpful was the result?

Scale: ⭐ 1 (Not useful) → ⭐⭐⭐⭐⭐ 5 (Extremely helpful)

Comments: ______________________________

Student 6: Michelle

Basic Information

Study Program: Business IT
Specialization: Project Management & Agility

Thesis Topic

Briefly describe your intended topic.

- Reputation signals in business school choice.

Research Question(s)

Write your main research question(s).

- What factors play a major role in choosing the right business school

Background Knowledge

How much do you already know about this topic?

☐ Nothing yet
☐ Some general understanding
☒Moderate knowledge (read some papers/articles)
☐ Deep knowledge (previous coursework, job experience)

(Optional: own personal experience and also work related.)

Goal of the AI Prompt

What do you want AI to help you with? (multiple allowed)

☐ Formulating or refining my research question
☐ Creating a structure or outline
☐ Defining methodology or data sources
☐ Writing summaries or theoretical background
☐ Clarifying concepts or terminology
☐ Generating visualizations or project plans
☐ Something else: ______________________________

Actual Prompt for AI

Write your best attempt at the prompt you'd like to give the AI.

I am writing my thesis about Reputation signals in business school choice. Give me a good research question on this topic .

Expected Outcome

What kind of result would you like AI to produce from your prompt?

☐ A refined research question
☐ A detailed outline or structure
☐ A summary or theoretical framework
☐ A list of datasets or tools
☐ A project plan (Gantt chart, milestones)
☐ Other: ______________________________

Feedback (optional)

After AI use: How helpful was the result?

Scale: ⭐ 1 (Not useful) → ⭐⭐⭐⭐⭐ 5 (Extremely helpful)

Comments: ______________________________

Student 7: Fereshteh

Basic Information

Study Program: Business IT
Specialization: Applied artificial intelligence

Thesis Topic

Briefly describe your intended topic.

AI-driven copyright compliance tools for software repositories.

Research Question(s)

Write your main research question(s).

Impact of AI copyright of training data on innovation in Switzerland

Background Knowledge

How much do you already know about this topic?

☐Nothing yet
☐Some general understanding
☒Moderate knowledge (read some papers/articles)
☐Deep knowledge (previous coursework, job experience)

Goal of the AI Prompt

What do you want AI to help you with? (multiple allowed)

☒Formulating or refining my research question
☐ Creating a structure or outline
☒Defining methodology or data sources
☐ Writing summaries or theoretical background
☐ Clarifying concepts or terminology
☐ Generating visualizations or project plans
☐ Something else: ______________________________

Actual Prompt for AI

Write your best attempt at the prompt you'd like to give the AI.

hey so you have access to 2 papers about AI and copyright I have a thesis similar project and I need to find a topic or a research question

….

I see but their research question was so simple and yet important, after looking at so many papers do you see any gaps that I could fill?

Expected Outcome

What kind of result would you like AI to produce from your prompt?

☐ A refined research question
☐ A detailed outline or structure
☐ A summary or theoretical framework
☐ A list of datasets or tools
☐ A project plan (Gantt chart, milestones)
☐ Other: _finding research gaps for potential research questions_____________________________

Feedback (optional)

After AI use: How helpful was the result?

Scale: ⭐ 1 (Not useful) → ⭐⭐⭐⭐⭐ 5 (Extremely helpful)

Comments: ______________________________

Student 8: Anna

Basic Information

Study Program: Business IT
Specialization: Applied artificial intelligence

Thesis Topic

Briefly describe your intended topic.

Agentic AI in Swiss financial services and bank industry

Research Question(s)

Write your main research question(s).

None yet

Background Knowledge

How much do you already know about this topic?

☐ Nothing yet
☒Some general understanding
☐ Moderate knowledge (read some papers/articles)
☐ Deep knowledge (previous coursework, job experience)

(Optional: briefly describe your current understanding.)

Goal of the AI Prompt

What do you want AI to help you with? (multiple allowed)

☒Formulating or refining my research question
☒Creating a structure or outline
☐ Defining methodology or data sources
☐ Writing summaries or theoretical background
☐ Clarifying concepts or terminology
☐ Generating visualizations or project plans
☐ Something else: ______________________________

Actual Prompt for AI

Write your best attempt at the prompt you'd like to give the AI.

I'm writing my bachelor's thesis on "Agentic AI in the Swiss Financial Services & Banking Industry." Help me define the research problem and main objectives that will be useful an realistic for me.

Expected Outcome

What kind of result would you like AI to produce from your prompt?

☐ A refined research question
☐ A detailed outline or structure
☐ A summary or theoretical framework
☐ A list of datasets or tools
☐ A project plan (Gantt chart, milestones)
☐ Other: _finding research gaps for potential research questions_____________________________

Feedback (optional)

After AI use: How helpful was the result?

Scale: ⭐ 1 (Not useful) → ⭐ ⭐ ⭐ ⭐ ⭐ 5 (Extremely helpful)

Comments: ______________________________

Student 9: Jimi

Basic Information

Study Program: Business IT
Specialization: Applied artificial intelligence

Thesis Topic

Briefly describe your intended topic.

The use of AI-based chatbots for mental health support

Research Question(s)

Write your main research question(s).

Background Knowledge

How much do you already know about this topic?

☐ Nothing yet
☐ Some general understanding
☐ Moderate knowledge (read some papers/articles)
☐ Deep knowledge (previous coursework, job experience)

(Optional: briefly describe your current understanding.)

Goal of the AI Prompt

What do you want AI to help you with? (multiple allowed)

☐ Formulating or refining my research question
☐ Creating a structure or outline
☐ Defining methodology or data sources
☐ Writing summaries or theoretical background
☐ Clarifying concepts or terminology
☐ Generating visualizations or project plans
☐ Something else: ______________________________

Actual Prompt for AI

Write your best attempt at the prompt you'd like to give the AI.

- I want to write my thesis on AI in psychology, with topic - the use of AI chatbots in mental health support. Please help me identify the main psychological concepts that are involved and help me with the scope of my thesis

Expected Outcome

What kind of result would you like AI to produce from your prompt?

☐ A refined research question
☐ A detailed outline or structure
☐ A summary or theoretical framework
☐ A list of datasets or tools

☐ A project plan (Gantt chart, milestones)
☐ Other: ______________________________

Feedback (optional)

After AI use: How helpful was the result?

Scale: ⭐ 1 (Not useful) → ⭐ ⭐ ⭐ ⭐ ⭐ 5 (Extremely helpful)

Comments: ______________________________

Expert Interviews

Interview 1: Léane Wettstein

In your experience, what makes it most difficult for students to choose an appropriate research methodology for their thesis topic?

Most of the time, students find this difficult because they lack experience and are not familiar with the different methodologies that are available. Because of this limited knowledge, it is hard for them to choose a method that is appropriate for their research question.

How often do you see students selecting methods that are unrealistic for their timeframe or available resources?
This happens quite often, but in different ways. Some students underestimate how much work a method requires and propose too little, so we have to tell them that a bachelor thesis needs a minimum amount of work. Others are too optimistic and include too many steps in their methodology.

Both situations occur frequently. It is very rare that a student already has a well-balanced method in their first draft. Usually, the method is either too ambitious or too limited. For example, some students propose conducting only four interviews for a bachelor thesis, but then we have to explain that they need at least twelve to generate sufficient insight.

When you talk with students, what concerns do they usually have about their thesis or thesis proposal?

Most of the time, students are unsure whether their research question is sufficiently scientific. They often worry that it might be too practical or not academic enough. This concern comes up very frequently.

Another major issue is the methodology. Students are often unsure whether their chosen method is appropriate for answering their research question, or whether they need to add or adjust certain elements. Generally, students understand their topic quite well, but they struggle more with formulating the research question and selecting the right methodology.

From your perspective, which aspects of the university guidelines are most confusing for students?
I cannot really answer this question because I am not familiar with the specific guidelines that students receive. I also do not know in detail what is taught in the different classes, so I cannot judge which parts might be confusing.

Are you familiar with AI tools, and have you noticed any patterns in how students use them for methodology-related tasks?
Yes. Students mostly use AI tools to summarize methodologies. For example, they find a method in a paper and then ask a chatbot to explain it or provide additional information. AI is also used to identify related papers that apply similar methodologies.

What limitations do you see in how students use these AI tools?
One limitation is that related methodologies often have very different names. If a student asks an AI tool about a specific method, it usually only provides information about that exact method. However, when reviewing the literature more broadly, students might discover that another related method is actually more suitable for their research question.

Another issue is inaccurate or false information. If students rely too heavily on incorrect AI outputs, this can lead to serious errors in their methodology section.

How should students verify whether AI-generated suggestions are correct or appropriate for a thesis?
Students should treat AI tools as a first step. Afterward, they need to verify the information using other sources. This could include reading scientific articles that explain the method in more detail or consulting research papers that apply it. Asking the supervisor is also an important step to ensure correctness.

Why do you think students rely on AI instead of seeking academic or methodological support from supervisors?
I think many students do not feel comfortable asking questions, or they feel that they do not have enough opportunities to ask them. Regular meetings with supervisors could help, but often students feel unsure about when or how to raise these issues.

From a lecturer’s perspective, what improvements would you suggest for the design of your chatbot prototype?
I would expect a place where important information about the thesis or project idea can be saved. This could be a history or dashboard showing previous discussions. Students will likely not work on everything in a single chat, so having access to different conversations would be important.

It would also be useful if students could enter their research question in a dedicated place, so the chatbot can consistently refer back to it when answering follow-up questions.

Should supervisors be connected to students through this app?
From a lecturer’s perspective, this could be useful, but not in the form of reading entire chat histories. Supervisors would need a concise summary of the student’s progress, such as an overview of research questions or a draft proposal template.

From a student's perspective, they might not want supervisors to see every question they ask. A summarized output would be more appropriate.

What kind of user input would be most useful for developing and evaluating this tool?
It would be helpful to understand what a student's first version of a research proposal looks like. This could be incomplete or rough, but it shows where students struggle the most. These real-world inputs can then be used to test whether the tool provides useful guidance.

Gold standards, in this case, would be accepted research proposals that were approved by supervisors. By analyzing several examples, you can identify common structures and ensure that the chatbot produces outputs that align with academic expectations.

INTERVIEW 2: LIVIA JOANNA MÜLLER

In your experience, what makes it most difficult for students to choose an appropriate research methodology for their thesis topic?

In general, methodology depends on what students want to find out. Many students are not equally comfortable with quantitative and qualitative approaches, and they sometimes choose a method that does not fit the question. For example, they may want to measure how satisfied people are, but then propose to "talk to them" in a way that does not actually answer that type of question.

Understanding what different methods can and cannot do is something that needs to be learned. In my experience, students struggle partly because institutions often lack hands-on courses that explicitly teach which research questions can be answered with which methodologies. In addition, students sometimes avoid certain methods simply because they feel uncomfortable using them.

How often do you see students selecting methods that are unrealistic for their timeframe or available resources?

This happens quite frequently, especially at the beginning of a thesis or project when students have limited project experience. Students often start with very ambitious ideas, but they struggle to break the project down into manageable steps and plan backwards from what they want to achieve.

In my experience, students tend to propose too much at the beginning. Later in the process, they often move to the opposite extreme and propose too little e.g., deciding to do only six interviews because they realize how much work is involved.

What concerns do students typically have about whether their research question is good enough or suitable for a thesis?

A common issue is balancing practical relevance with academic contribution. In some contexts, students work with industry partners. Industry partners often want something that is practically relevant, but academically it may not contribute much. However, a thesis typically needs to include both: practical value and some contribution to theoretical knowledge.

Another challenge is that students sometimes propose ideas that already exist because they have not done enough reading. A solid research question requires understanding the field: what has already been done and what recent developments are. If students rely only on a superficial summary (for example, from an LLM), their view of the field can remain incomplete.

Students also struggle with making the research question answerable. They may start with an abstract question like “How can I make my users happier?”, but then it becomes unclear how to measure that and what concrete steps and methods would be needed to answer the question. Moving from an abstract problem to a concrete, operationalizable research question is difficult and happens frequently.

From your perspective, which aspects of the university guidelines are most confusing for students?

I cannot answer that in detail because I do not know the guidelines by heart, at least not the ones from this institution.

When students try to narrow their focus, what challenges do you observe?

I would describe this as a “zooming in and zooming out” issue. At one level, students need to zoom out and decide what is relevant for the thesis: which theories, concepts, or elements actually matter for the argument. They need to check whether the logic and overall structure make sense.

At the same time, they need to zoom in and evaluate whether the specific argument they are making is valid. For example, if they claim they are researching user happiness by using a survey, they need to check whether this method actually captures what they want to measure and whether it truly answers the research question. This repeated switching between the big picture and the details is a major challenge.

Are you familiar with AI tools, and have you noticed patterns in how students use them during the thesis process?

Yes. I encourage students to use these tools, because they can be very helpful and can speed up parts of the process. However, students need to use them critically and reflectively.

A common problem is that students copy AI-generated text without understanding it. In thesis supervision, it is often obvious when a sentence has been taken from an LLM because it does not clearly mean anything or the student cannot explain it. For example, I have seen students label something as a “study” in a title when it was actually only a literature review. When asked to explain what they mean, they cannot.

AI tools can support idea generation and refinement, but they should not replace thinking. If students use AI without reflection, they may complete a degree without developing real understanding, which creates problems later in professional work.

Why do you think students go to AI tools instead of asking supervisors for help?

One reason is that students often do not know where to start or what to ask. In that situation, even a random AI output can help them begin thinking and clarify what they are actually struggling with. It can provide an initial input that helps them see the problem from different perspectives and eventually form more precise questions.

Being able to ask for help clearly is also a skill: identifying the problem, specifying what kind of support is needed, and requesting feedback on a specific part of the work. Students may not have developed this skill yet.

In addition, asking supervisors can feel intimidating. Some students fear feedback or worry they will appear incompetent. In that sense, AI can feel like a safer first step to explore ideas before approaching a supervisor.

Could you give an example of what a "good proposal" (gold standard) looks like?

A strong proposal typically shows clear thinking and preparation. For example, I recently supervised two students who already had a concept from a seminar and were very clear about the problem they wanted to address.

Their proposal had a clear problem statement and a clear solution. The problem was described concretely (including who is affected and what the consequences are). Then the solution was explained clearly, including why it was designed in that way. The methodology followed naturally from the problem and solution, and the proposal was structured step by step: problem, solution, how it will be implemented, and who benefits.

That clarity and logical flow are what I value most in proposals.

INTERVIEW 3: MATTHIAS SAHLI

What are the most common difficulties you observe when students begin writing their thesis proposal?
That's clearly the research question. I think most students struggle with identifying a precise and well-defined research question. Usually, they formulate it too broadly, and they suffer later on if they don't have a clear and focused question.

How do you usually help students define their research question and methodology?
Finding a good research question requires a lot of reading. Typically, interesting and relevant questions develop during that process. I know it takes time, but in order to formulate a solid research question, you need to understand the area of research you're entering. My approach is to help students find interesting readings first, then schedule a meeting to discuss possible research questions and work on the methodology from there.

When a student asks for help with a research question or method (like in these examples), what would you consider a good or successful outcome of your support?
To motivate students by providing them with interesting readings that help them build a strong foundation.

What do you think a “gold standard” AI-supported proposal draft should achieve?
What you shouldn’t do is ask an LLM very open questions. Ideally, use it in the early stages to be creative. Afterwards, it’s your job to put together interesting and high-quality content. That’s where I see a lot of potential in writing support.

What characteristics distinguish a well-formulated research question or proposal section (e.g., clarity, feasibility, alignment with methodology, relevance)?
All of them a good research question should have all these characteristics.

If an AI assistant were to support students in this process, what tasks should it ideally handle and which should remain human-driven?
My feeling at the moment is that it’s not very helpful for identifying and gathering good papers or sources for a literature review at least for me personally, it hasn’t worked well. Where it’s always useful, though, is with language and writing support.

What kind of prompts or questions should AI ask students to guide them effectively?
The LLM shouldn’t ask open questions. It should help you narrow down your direction and assess feasibility for example, would it be possible to gather data in this field? Will you have access to the required experts?

How could AI tools best support method selection (qualitative vs. quantitative, design science, etc.)?
That really depends on what input the student provides.

What potential risks or misconceptions do you see if students rely too much on AI for their proposal?
Clearly, improper citations. Drafts should always be checked carefully.

What indicators would you use to compare an AI-generated output to a student’s own draft?
AI-generated texts tend to be too long, use too many bullet points, lack relevance, and are not concise or well-focused. Citations are also often missing or incorrect.

Could you provide examples of prompts or interactions that would produce high-quality results in your view?
Off the top of my head let’s say you already have a draft of your proposal. You could give it to an LLM and ask:

“Ask me five tough questions and challenge my research proposal as if you were my professor, so I can identify weaknesses before submission.”

Can you give us some good proposals that you have received from a student?
From a data-compliance perspective, I'm not sure I'm allowed to do that. I would need to check I'm not permitted to share such proposals directly.

INTERVIEW 4: LUCIA GOMEZ

What are the most common difficulties you observe when students begin writing their thesis proposal?

To be concise. The same problem I observe in students is what I observe in AI systems: when a student is exploring a topic for the first time, topics are chosen but the student isn't an expert. The ideation becomes too broad, ideas go all over the place, and there's a lack of realistic thinking about what's measurable. Questions are too broad and hard to measure. Instead of starting from "what can I do and how can I formulate a question that fits what I can do," there's "that would be awesome to answer," but it goes beyond a three- or four-month piece of work.

How do you usually help students define their research question and methodology?

Think about methodology first what would actually be done. From there, realize what's possible and formulate the question bottom-up. Consider what you can do in this three-month window, your abilities, abilities you want to develop, and the skills you want to leverage. Put hypotheses you can formulate and measure together, and let the overarching question emerge.

So exactly how to answer the methodology itself?

Yes, ask what you can do realistically within the time and skills you have. Around the topic that interests you originally very broad think what you can do and *how* you would do it conceptually (e.g., interviews, web scraping). Your method shapes what you can answer; formulate the question you can answer with the method you can execute.

When a student asks for help with a research question or method, what's a successful outcome of your support?

Progress, the student shouldn't feel overwhelmed or afraid to try. It's hard to know the ideal support. Empathy matters, but avoid codependency. I share my perspective and let the student decide. If I don't see progress and the student is stuck, I re-examine my approach: "Lucia, you're not doing a good job now how can you do better?"

What should a "gold-standard" AI-supported first draft achieve or include?

It should include what BFH requires: title, a clear question, hypotheses and how to test them, why it's important (the gap). The order: **importance → gap → question → steps/hypotheses to test**. The AI should help students *think like researchers* and go through that process. If the AI does it *for* them, they won't learn and will struggle later. Developing scientific thinking takes time; a supervisor has limited hours, so the agent should foster that thinking and provide critical feedback to deepen reasoning, not replace it.

What tasks should the AI ideally handle, and what should remain human-driven?

The agent should act like a supervisor with endless time teaching scientific thinking and guiding critically. The *student's perspective* must remain human. If the agent produces the proposal, the student won't "click" with the thesis; the problem persists later.

So it shouldn't just give a completed thesis proposal; it should guide students to write it themselves.

Absolutely. If someone always does things for you, you don't learn. The AI should challenge you. My PhD advisor often said "not convincing go back." You need to *convince* a strict standard. The agent could have modes: friendly, spicy, and killer. Personally I'd choose killer mode, to ensure minimum quality. I write a rough draft, then iterate with AI section by section, asking it to restructure or question me but the content starts with me. The student must keep ownership of the story; AI should highlight what the student wants, not impose a plan that misaligns with them. Otherwise you could end up over-promising something you don't understand.

What questions should AI ask students to help them?

Use the structure to guide thinking one question after another and then add "inside the replies" questions:

- Why this topic? Why is it important?
- Are you sure it's important? What makes you think that?
- Is it specific enough?
- Could you explain it to your mother or grandmother? If they say "who cares?" you haven't found the sweet spot.

How could AI tools support method selection?

AI is useful to survey methods once you have specific hypotheses and data. If you say, "I have this type of data and can use Excel/Python/SPSS/by hand," AI can list suitable methods. That's faster than searching the web blindly.

What risks or misconceptions arise if students rely too much on AI in their proposal?

They won't develop critical/scientific thinking. A proposal forces you to create a mental map a sequence of steps toward the goal over months. If AI makes that map, you miss the struggle and learning; problems reappear later.

What indicators make you think "this is AI-generated"?

Broad, generic phrases that could mean many things at once. Weird, AI-ish words.
If it's AI-style *but specific*, that's okay suggests iteration. If it's AI-style *and broad*, it's not ready.

Some students who avoid AI struggle with structure (importance → gap → question → method), causing disordered narratives; AI respects structure but can be empty.

Can you provide examples of prompts/interactions that produce high-quality AI outputs?

1. Student writes a brainstormed draft.
2. Insert square-bracket placeholders like [fill this with a specific…].
3. Ask the AI to fill only placeholders and to **bold** anything it adds/changes.
4. Iterate: highlight parts and say "not convincing find references."
   This maintains transparency about AI vs. student content.

So the student writes a draft, leaves placeholders, and tells the LLM to bold its own contributions.
Yeah.

Any essential feedback mechanism or design principle for the tool?

The role should be *supervisor*, not student or researcher. It should prioritize sustainable skill development, not dependency. Include configurable **criticism levels** (friendly → unconvincible). The AI should critique our outputs like we critique AIs. Possibly include modes: supervisor, helper, creative idea.

It would be nice for students to say: Enter supervisor mode / helper mode / creative idea mode.
That would be hard, but great.

Could you share proposal examples we could use?

I can ask my past students for permission. In Moodle, they're not open. I'll pick the best from last semester and send them. I'll contact as many students as I can. But examples aren't strictly necessary. You need steps that make students think and fill sections. Since topics vary, you don't want the AI to anchor to specific domains.

If not BFH proposals, what examples would be better?

Use **pre-registered studies** (e.g., on the **Open Science Framework**). They're like proposals written by experienced researchers; you can't change hypotheses or methods after registration, so they're robust. These make stronger exemplars than student proposals.

Interview 4: Sebastian Höhn

What are the most common difficulties you observe when students begin writing their thesis proposal?

Honestly, it's almost always the research question. Students tend to start way too broad. It sounds interesting at first, but later on they realize it's hard to handle because it's not really focused. That usually causes trouble down the line.

How do you usually help students define their research question and methodology?

I usually tell them: read first, think later. A good research question comes from understanding the field. So I point them to some key papers, let them explore a bit, and then we sit down and talk through what might actually work and which methods would make sense.

When a student asks for help with a research question or method, what would you consider a good outcome?

For me, it's a good outcome when the student feels less stuck and more confident. If they walk away with a clearer direction and a few solid readings that help them understand what they're doing, that's already a win.

What do you think a "gold standard" AI-supported proposal draft should achieve?

AI shouldn't write the proposal for you. It's much better as a thinking tool, especially early on. It can help generate ideas or improve wording, but the real work deciding what's relevant and making it academically sound still has to come from the student.

What makes a research question or proposal section really good?

There's no single magic trick. It just needs to be clear, doable, and aligned with the method. And of course, it should actually matter for the field. If all of that fits together, you're usually in good shape.

What should AI handle, and what should stay human?

I don't think AI is great at finding or judging academic sources at least not yet. That still needs human judgment. Where it really helps is with writing: making things clearer, cleaner, and easier to read.

What kind of questions should AI ask students?

Not very open ones. It should help narrow things down. For example, it could ask whether the data is actually available, or whether the student has access to the right people or cases. Those practical questions are really important.

How can AI support method selection?

That depends a lot on what the student puts in. If the input is clear, AI can suggest reasonable options. If it's vague, the output will be vague too. It's not magic it works with what it gets.

What risks do you see if students rely too much on AI?

Citations, without a doubt. AI can sound very confident while being completely wrong about references. So students really need to double-check everything before submitting anything.

How can you tell an AI-generated text from a student's own work?

AI texts often feel a bit bloated. They're long, full of bullet points, and not very focused. They also tend to be generic. Student-written drafts usually feel more specific and to the point even if they're not perfect.

Can you give an example of a good AI prompt?

One I really like is asking the AI to be critical instead of helpful. For example, Act like my supervisor and challenge this proposal. What are the weak points? That can be surprisingly useful.

Can you share examples of strong student proposals?

Unfortunately, no. mostly because of confidentiality and data protection rules. Even really good proposals usually can't be shared without permission.